



U.S. Flight Delays

2024 - 2025

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Agenda



1 Introduction

2 Problem Statement

3 Objectives

4 Analysis

5 Recommendations





Introduction

Air travel has become an essential part of modern life, **connecting people, businesses, and economies worldwide**. As **air traffic grows**, maintaining **efficient and timely operations** becomes **increasingly important**. Flight delays can affect **passengers, aircraft** and **crew schedules**.

The **Bureau of Transportation Statistics (BTS)** supports greater understanding of U.S. air transportation by **collecting and providing data on its operational performance**. These insights are particularly relevant to **Airline Operations Managers** and **Airport Operations Managers**, supporting the monitoring of **operational performance**, and **identification of delay patterns**.

Problem Statement

During **2024–2025**, approximately **26.8%** of **reported U.S. flights** experienced at least one **official delay of 15 minutes or more (departure or arrival)**. These delays **reduce schedule reliability, create operational challenges for airlines, and increase costs on both passengers and airlines.**

Objectives



Flights Overview

Provide an overview of **flight volume**, **trends**, **airline distribution**, and **geographic distribution** across **states**, **cities**, and **airports**.



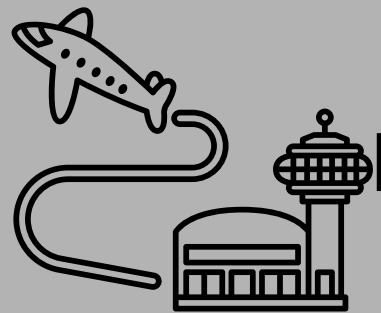
Flights Delays

Identify the **number and percentage of delays**, and highlight **trends** and distribution across **airlines** and **locations**.



Delay Severity & Causes

Assess **delay severity** by **duration** and identify the distribution of **moderate, and major delays**, and the contribution of **delay causes in minutes**.



Airport-Level Delays

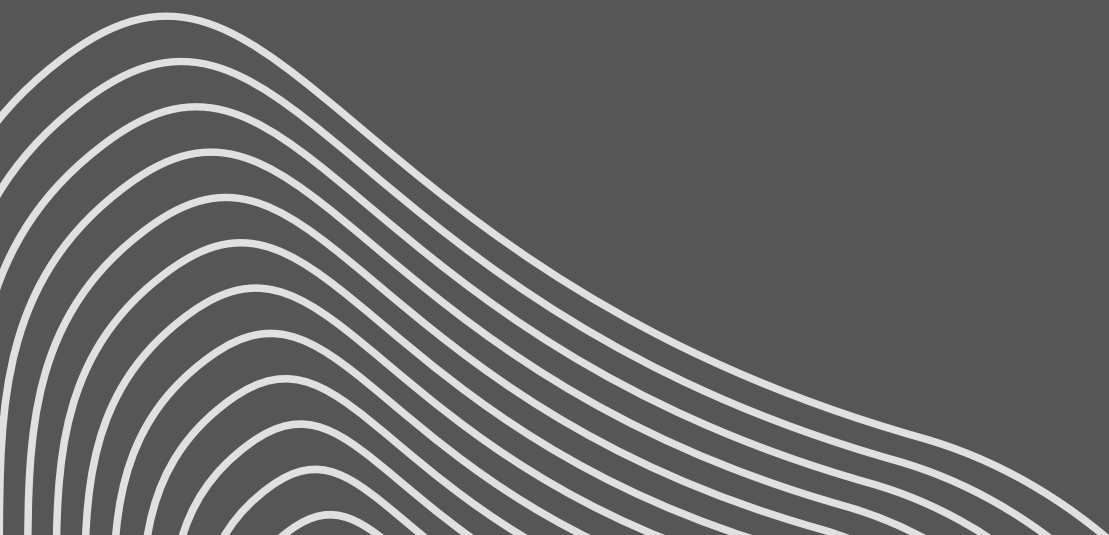
Focus on a **high-delay airport** to identify its **flight volume, delay patterns, severity, trends, and key contributing causes**.

Analysis





Flights Overview



Flights Overview

Time Frame

2024
2025

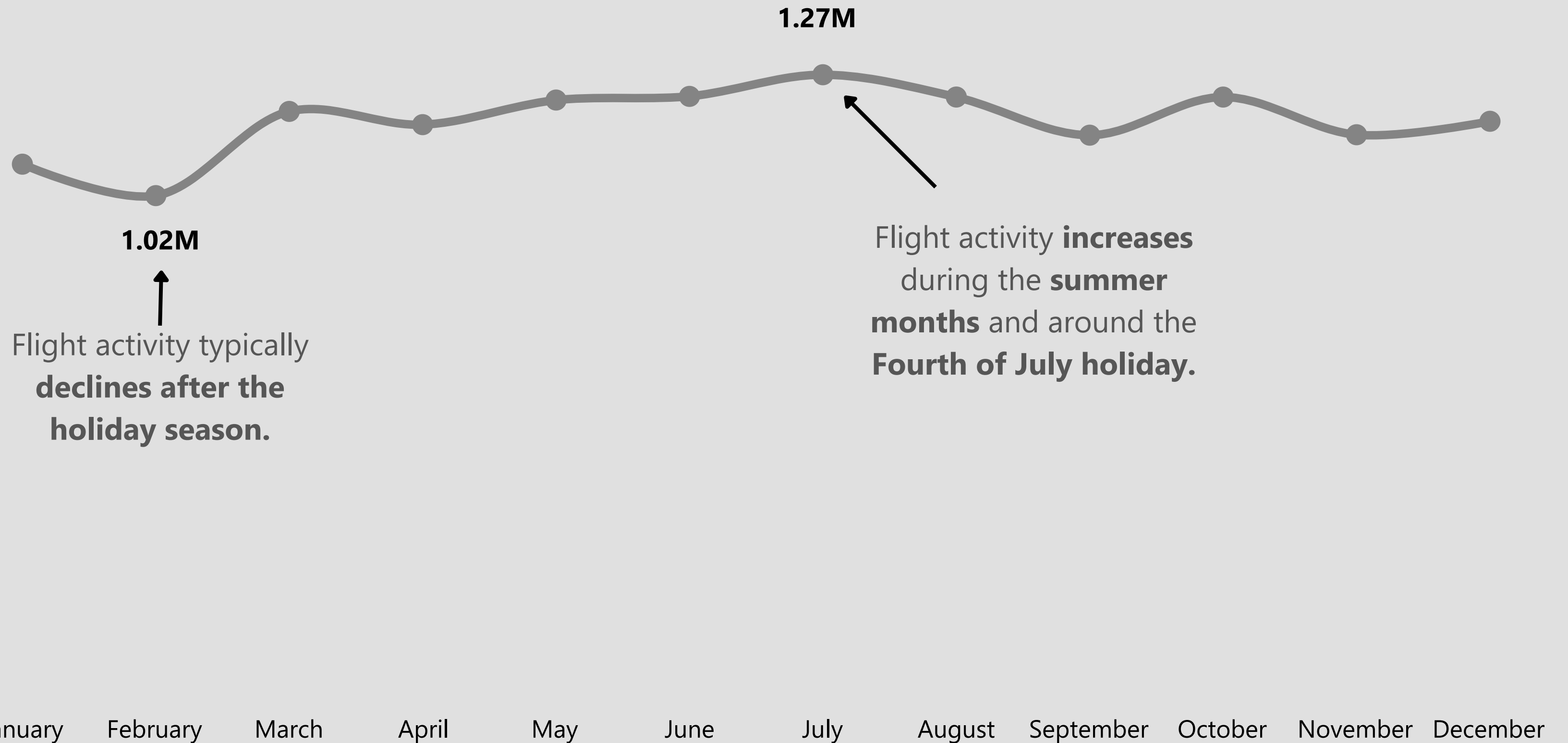
Total Flights

14.08 M

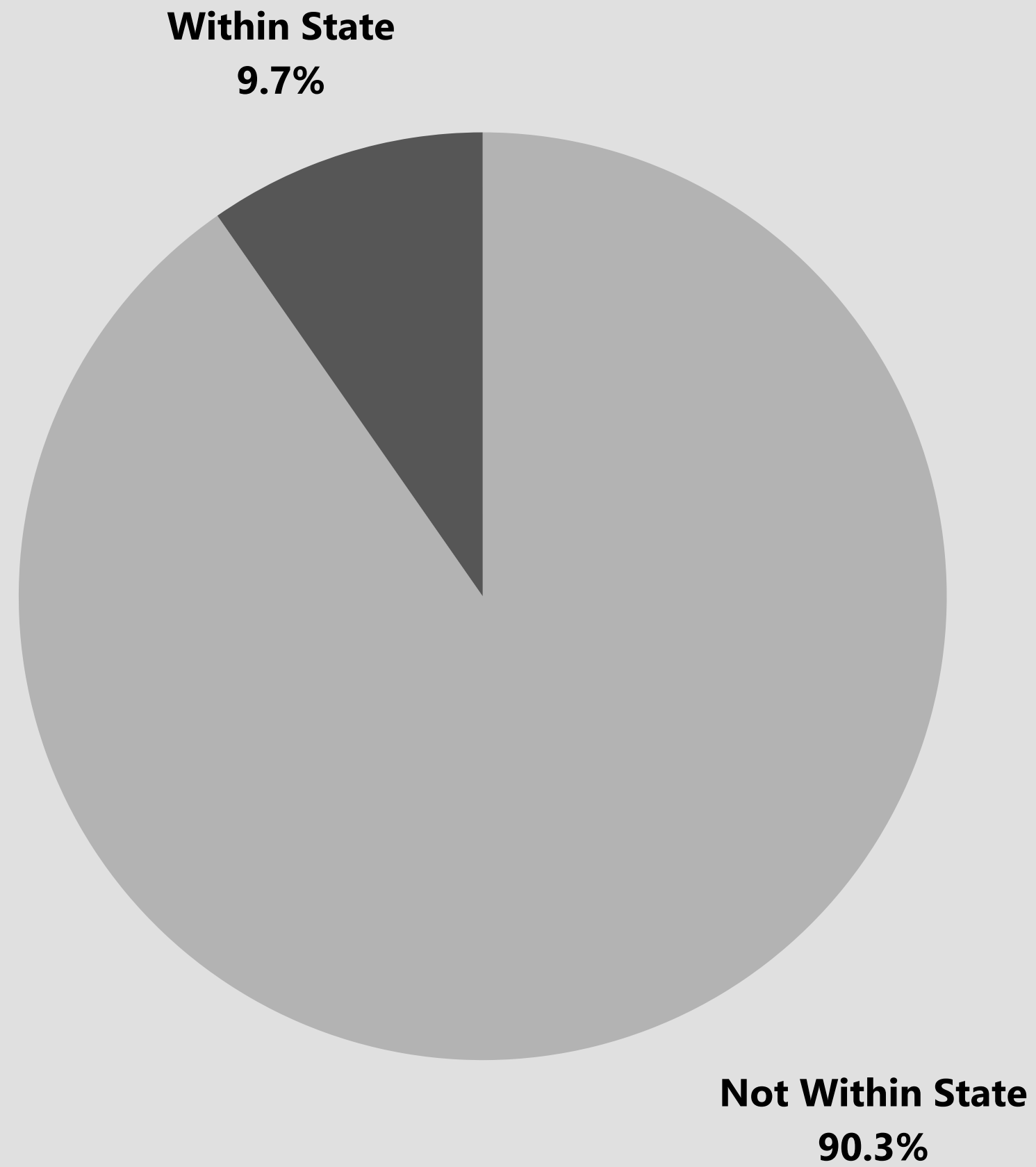
Number of Airlines

15

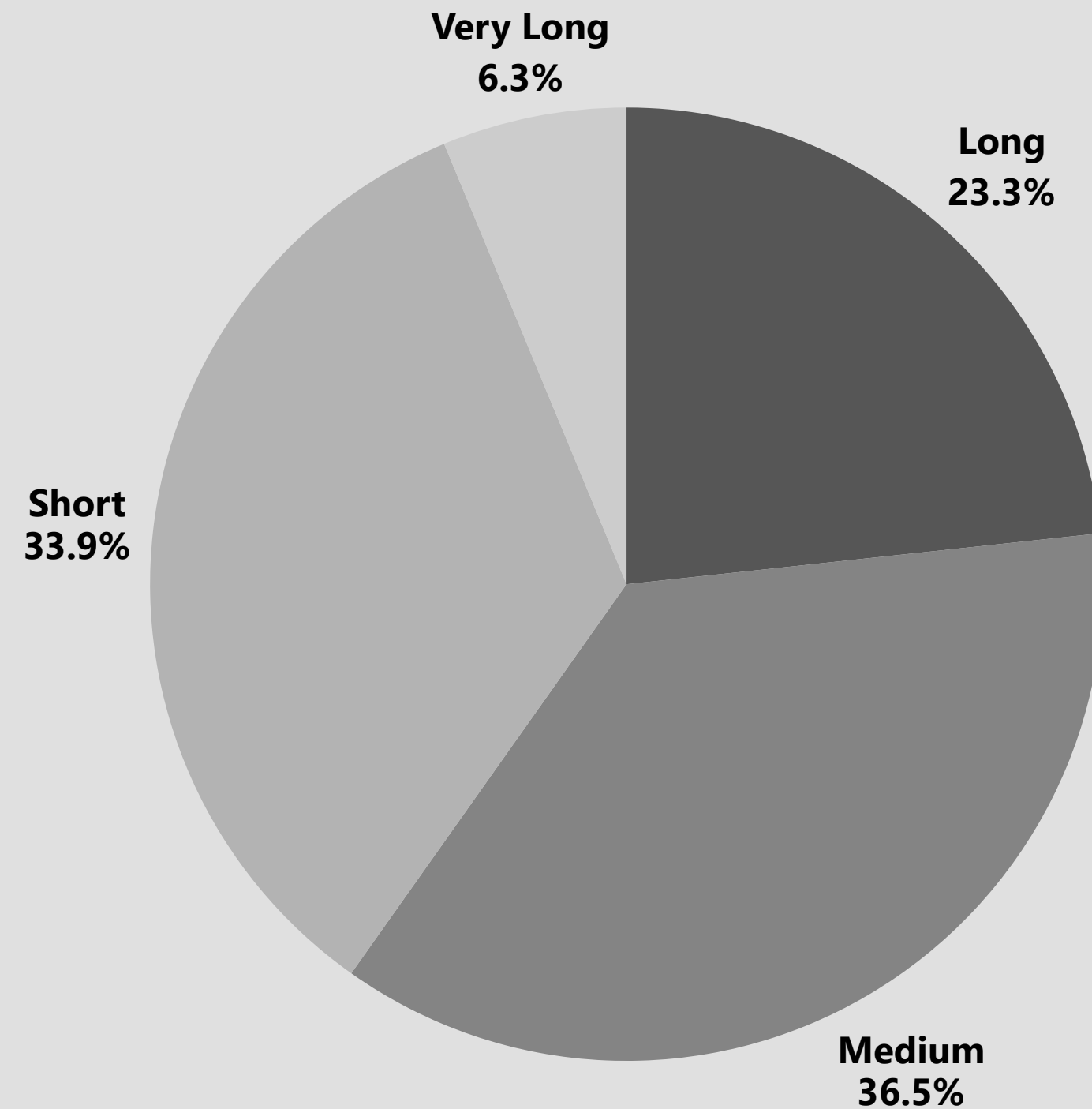
Number of Flights during the Year



Percentage of Flights within a State



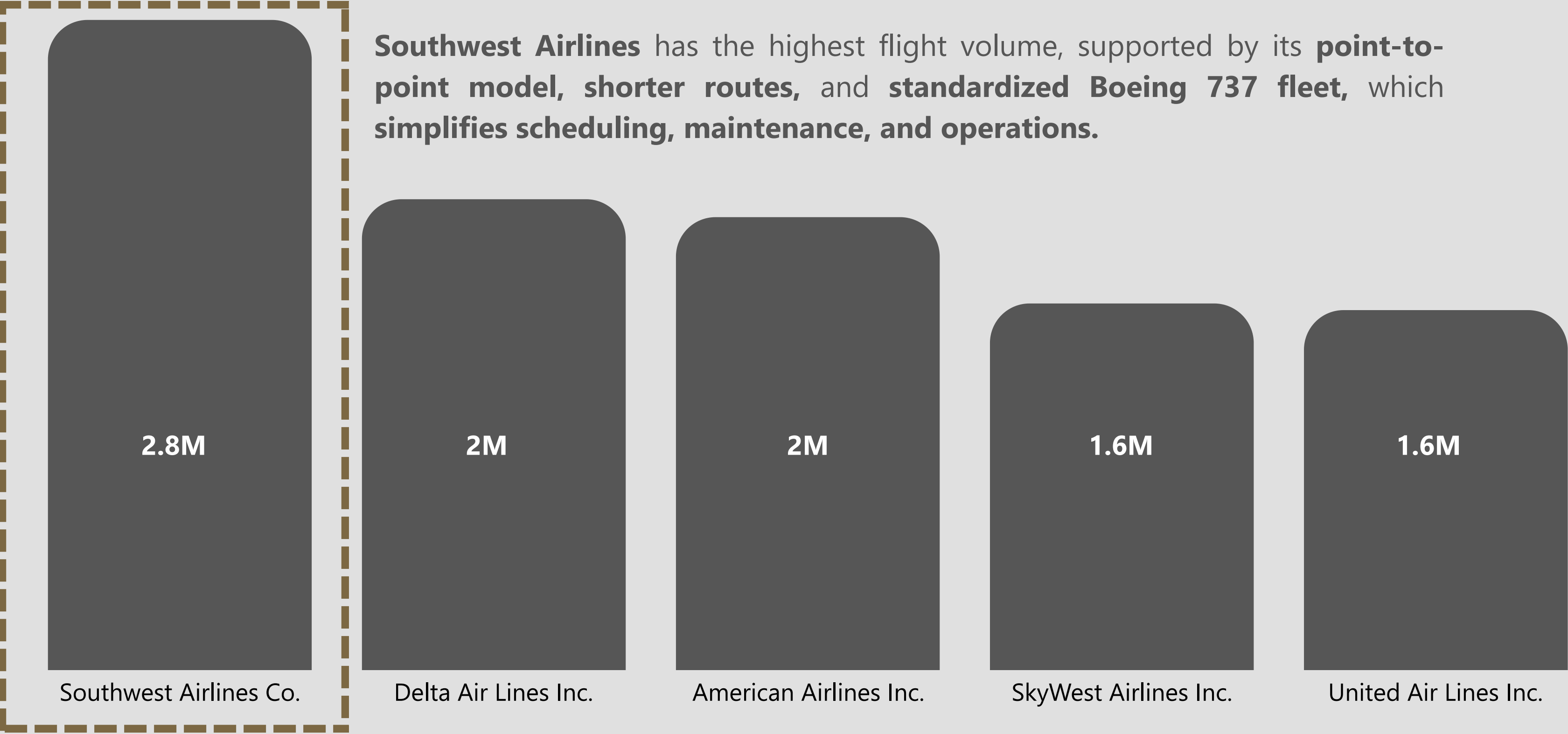
Distance of Flights



Distance Categories

- Short: < 500 miles
- Medium: 500–999 miles
- Long: 1,000–1,999 miles
- Very Long: 2,000+ miles

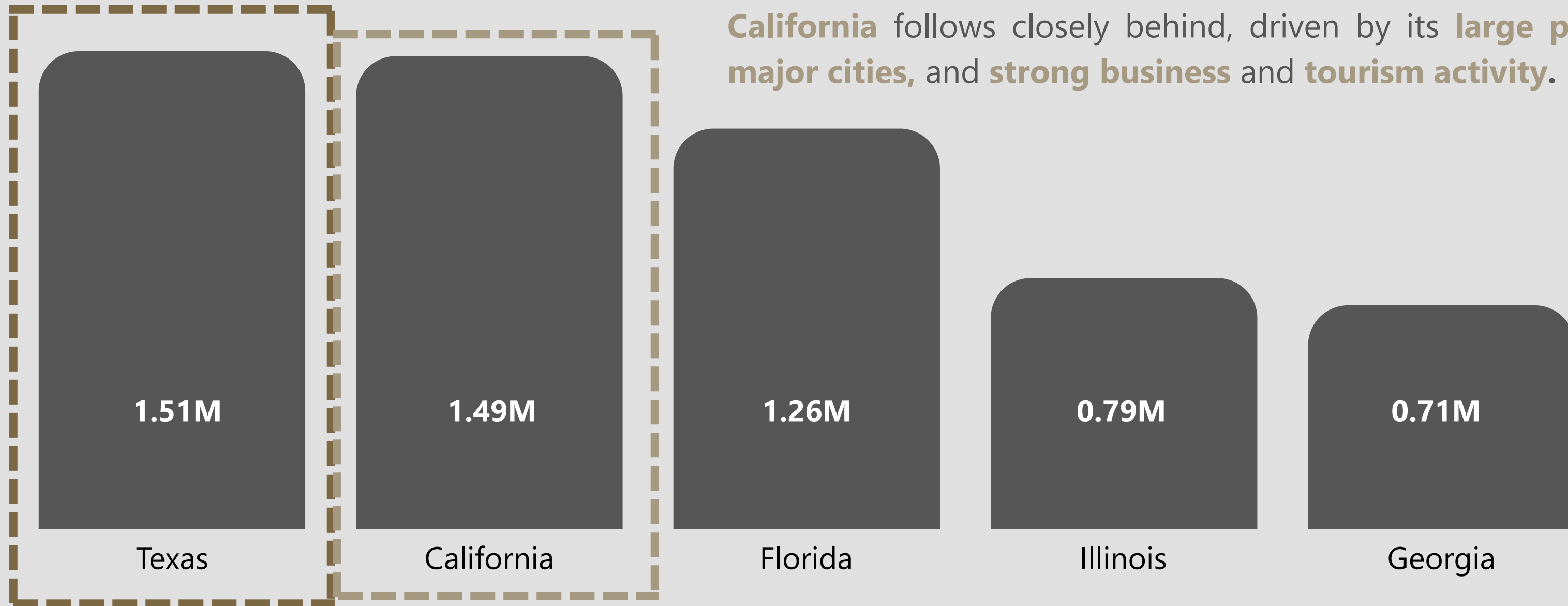
Top 5 Airlines by Number of Flights



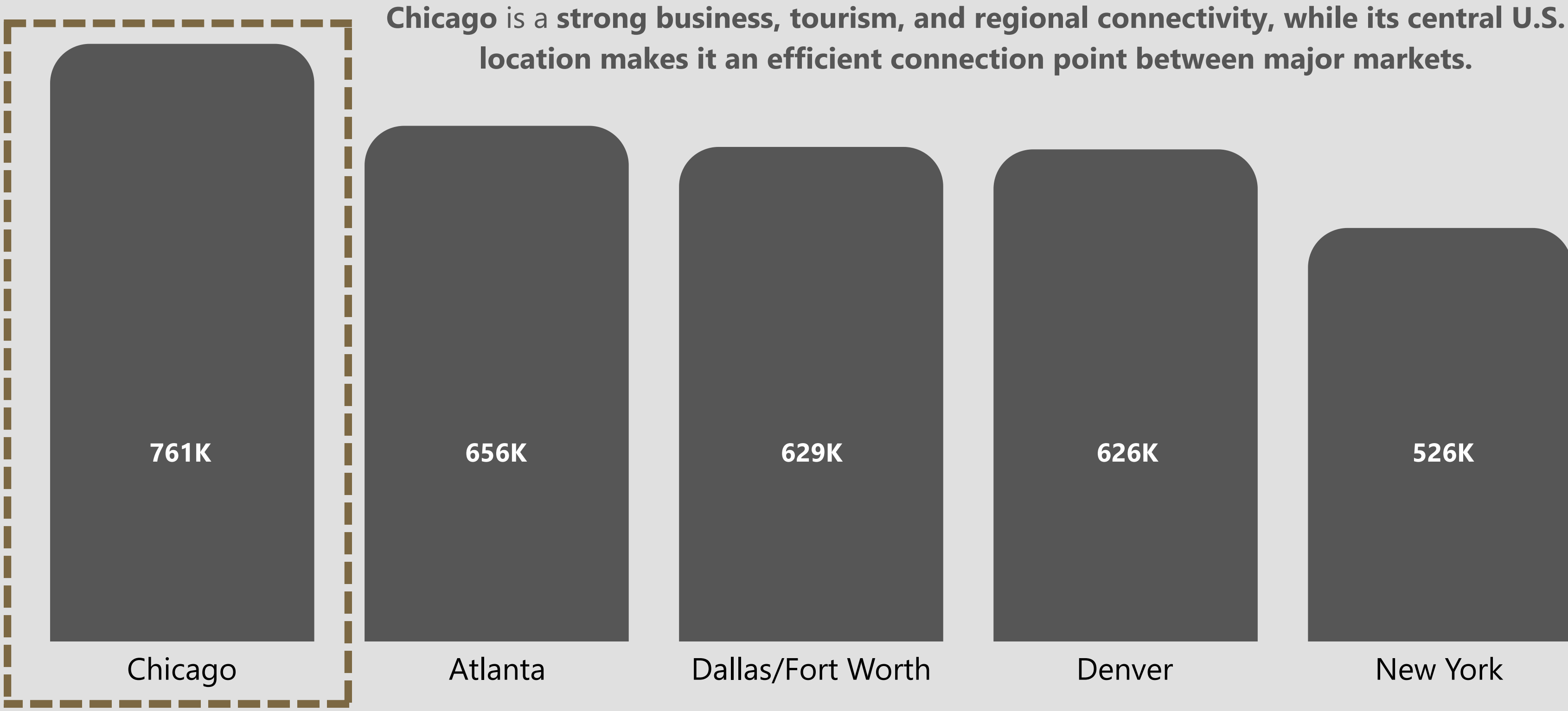
Top 5 States by Number of Flights

Texas has the highest flight volume because its **large population, strong economic activity, and extensive transportation network** create **high demand for air travel**.

California follows closely behind, driven by its **large population, major cities, and strong business and tourism activity**.

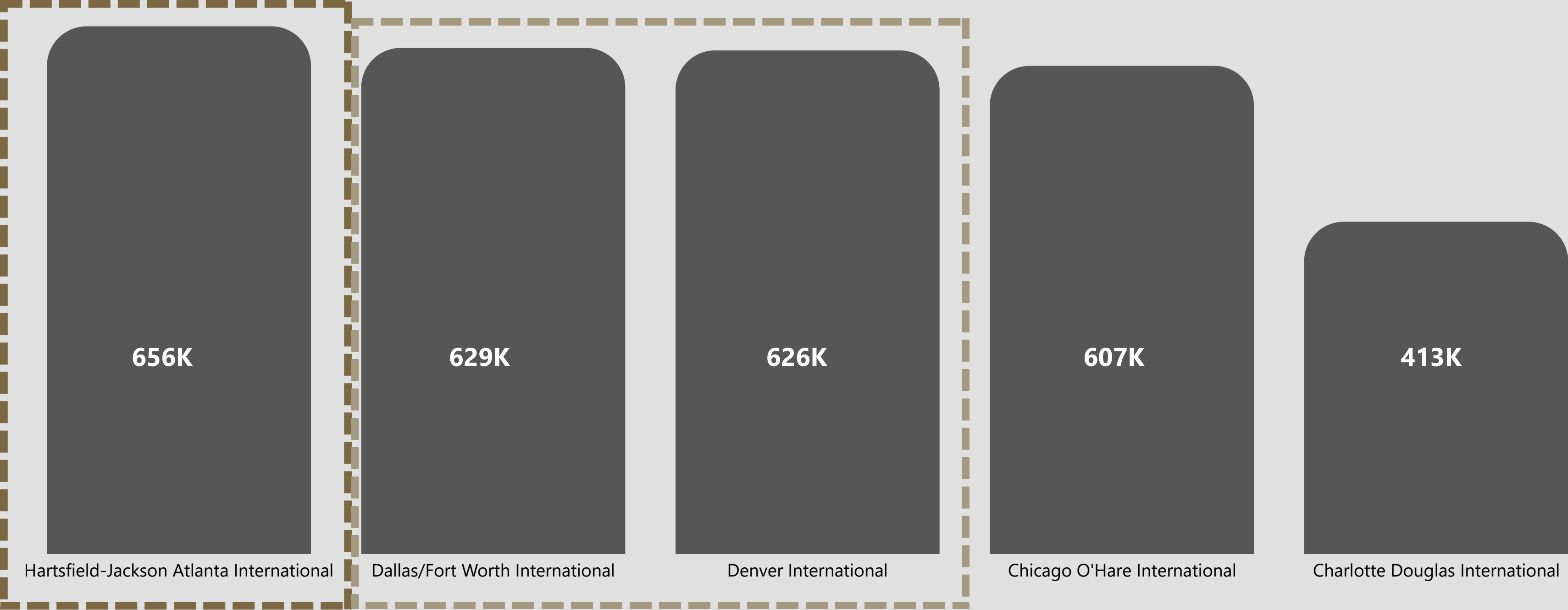


Top 5 Cities by Number of Flights

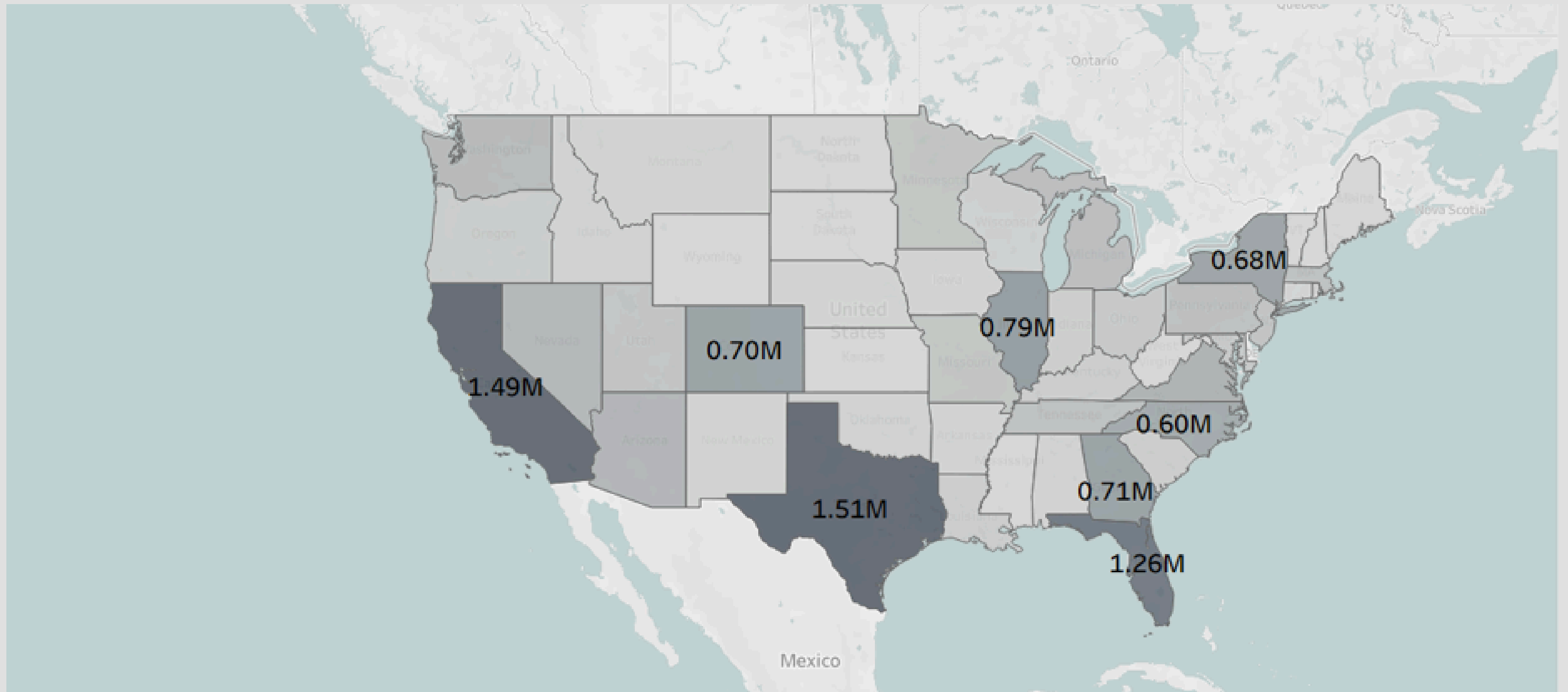


Top 5 Airports by Number of Flights

Atlanta has the **highest flight volume**, followed closely by **Dallas/Fort Worth** and **Denver**, due to their roles as major connecting hubs. Atlanta, DFW, and Denver serve as major hubs for **Delta Air Lines**, **American Airlines**, and **United Airlines**, respectively, generating substantial connecting traffic.

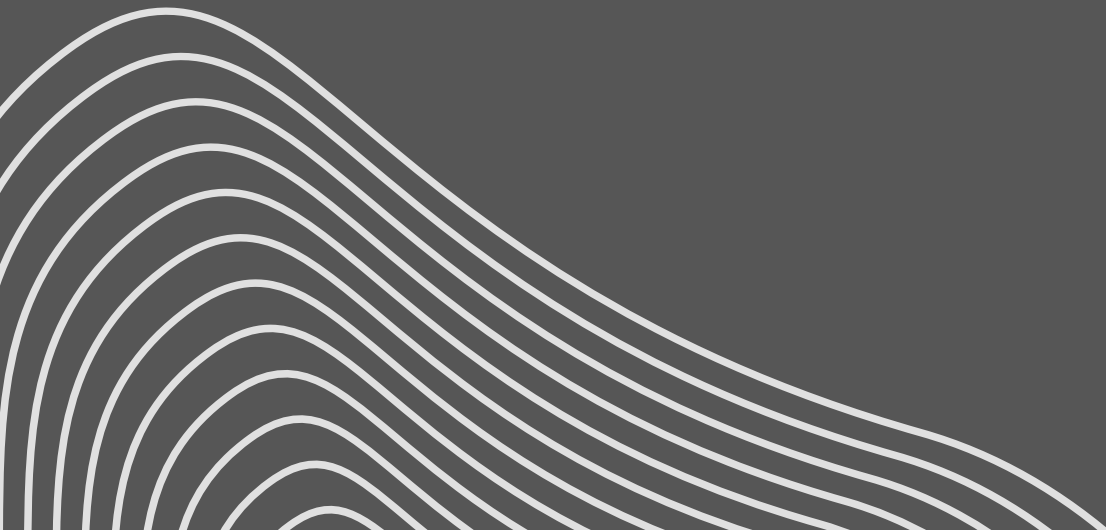


Top 5 Airports by Number of Flights





Delay Analysis



ON-TIME PERFORMANCE (OTP)



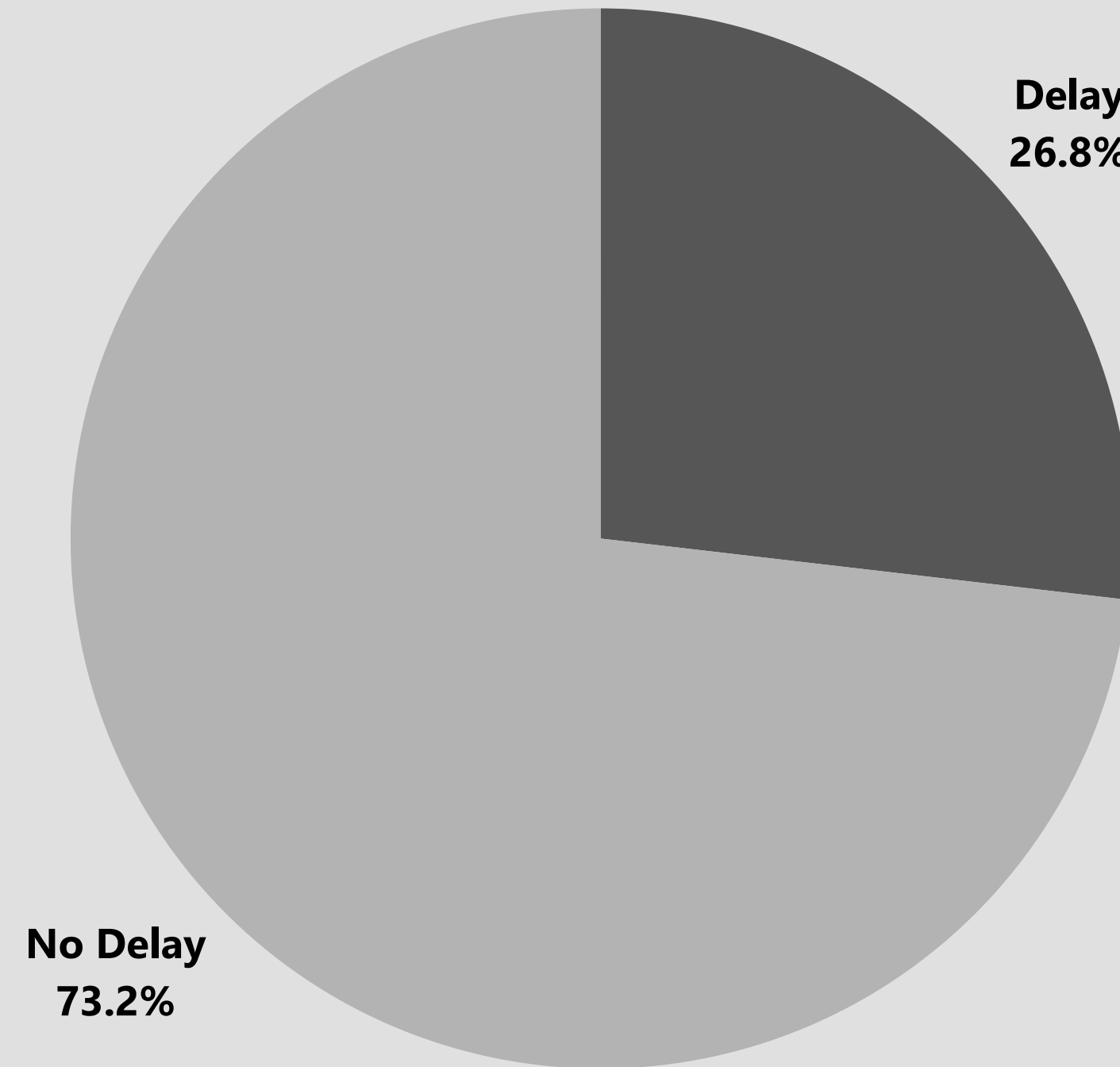
A flight is considered “**on time**” when it arrives at the scheduled destination within **15 minutes** of its scheduled arrival time.

OTP = A key measure of airline and airport operational performance.

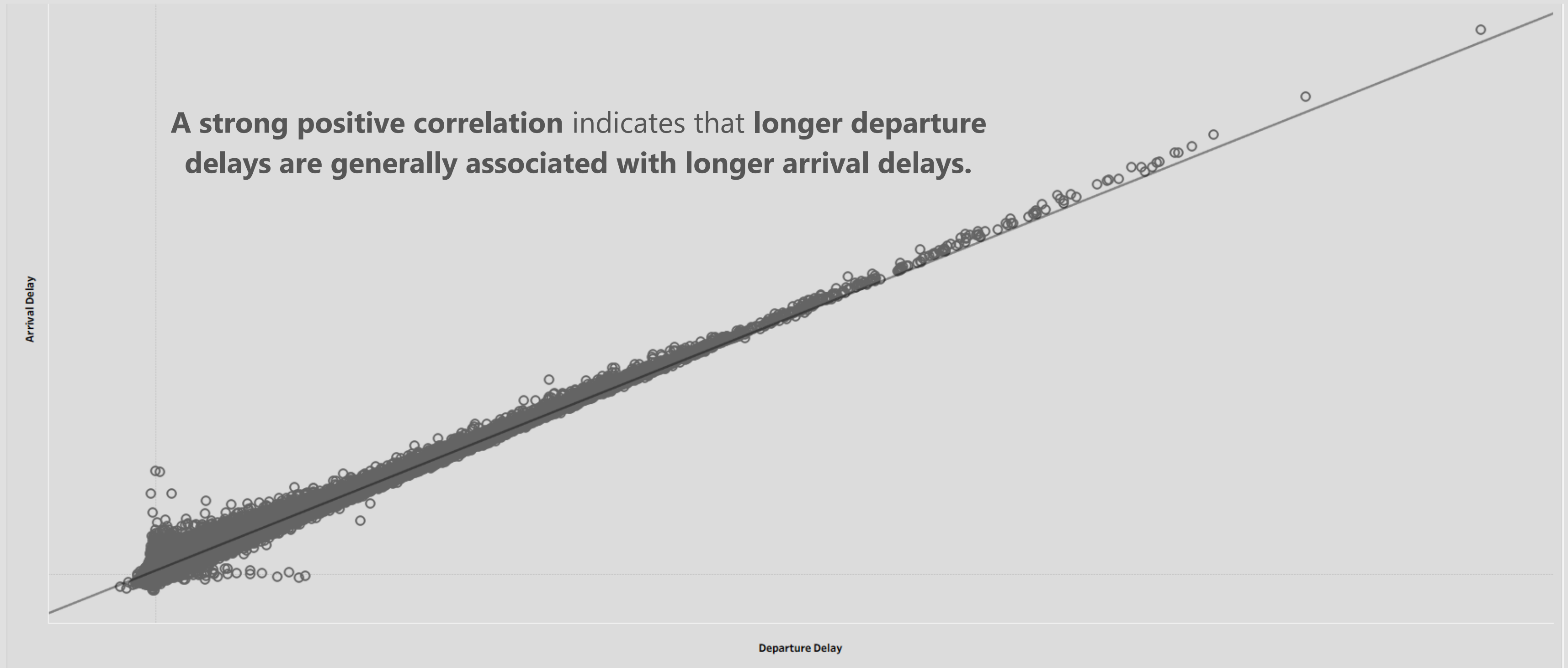
Source: Bureau of Transportation Statistics



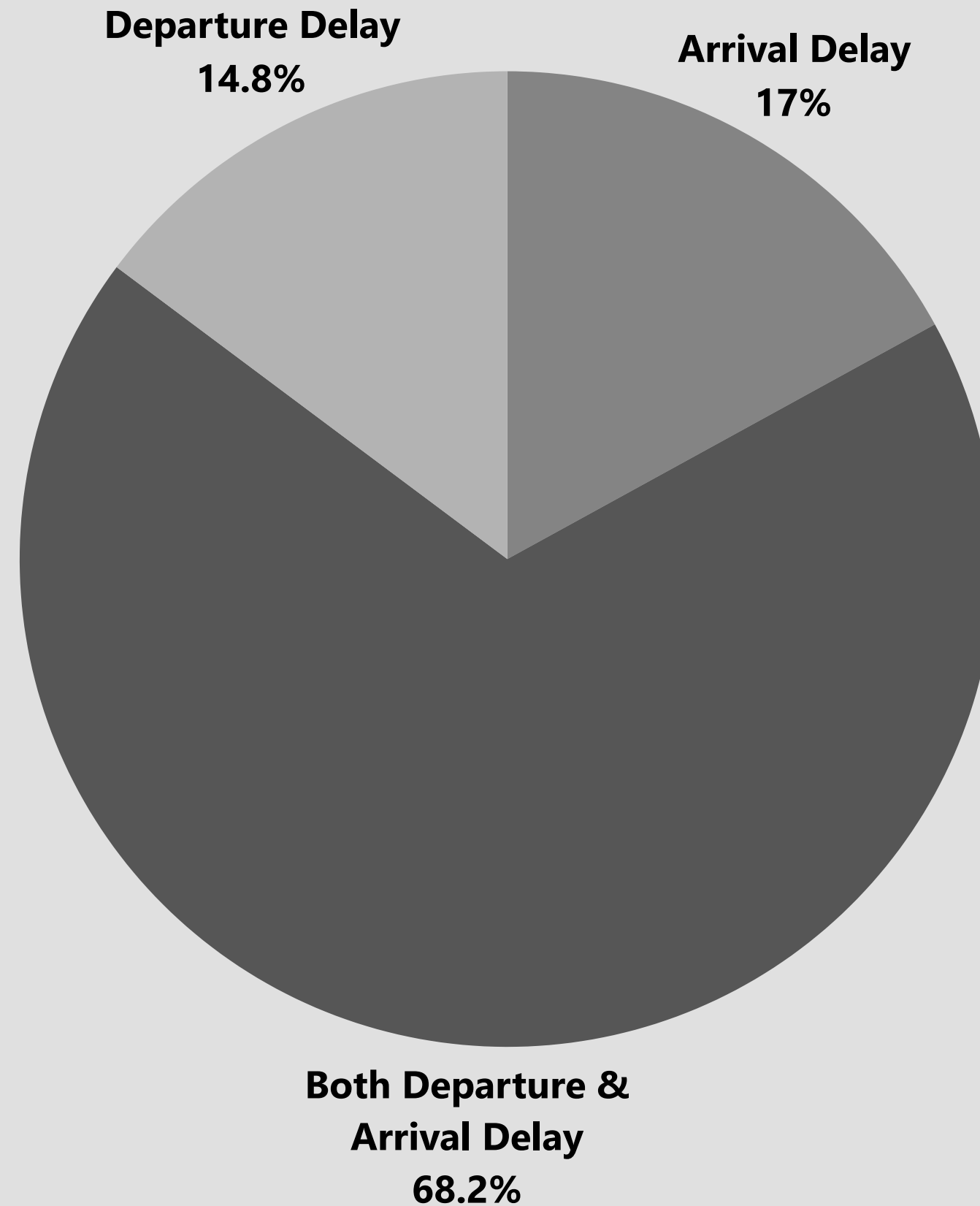
Percentage of Delayed Flights



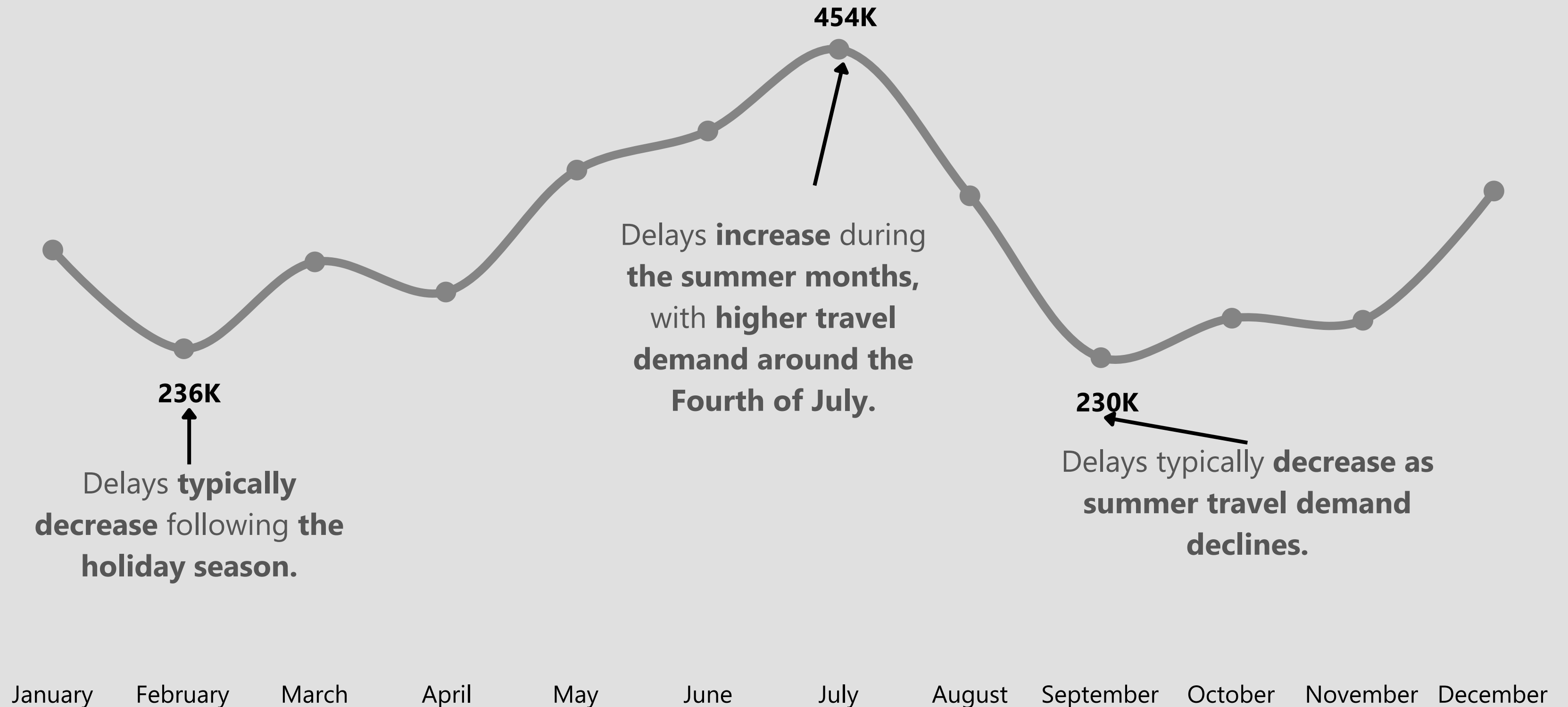
Correlation between Arrival & Departure Delay



Percentage of Delayed Flights by Delay Type

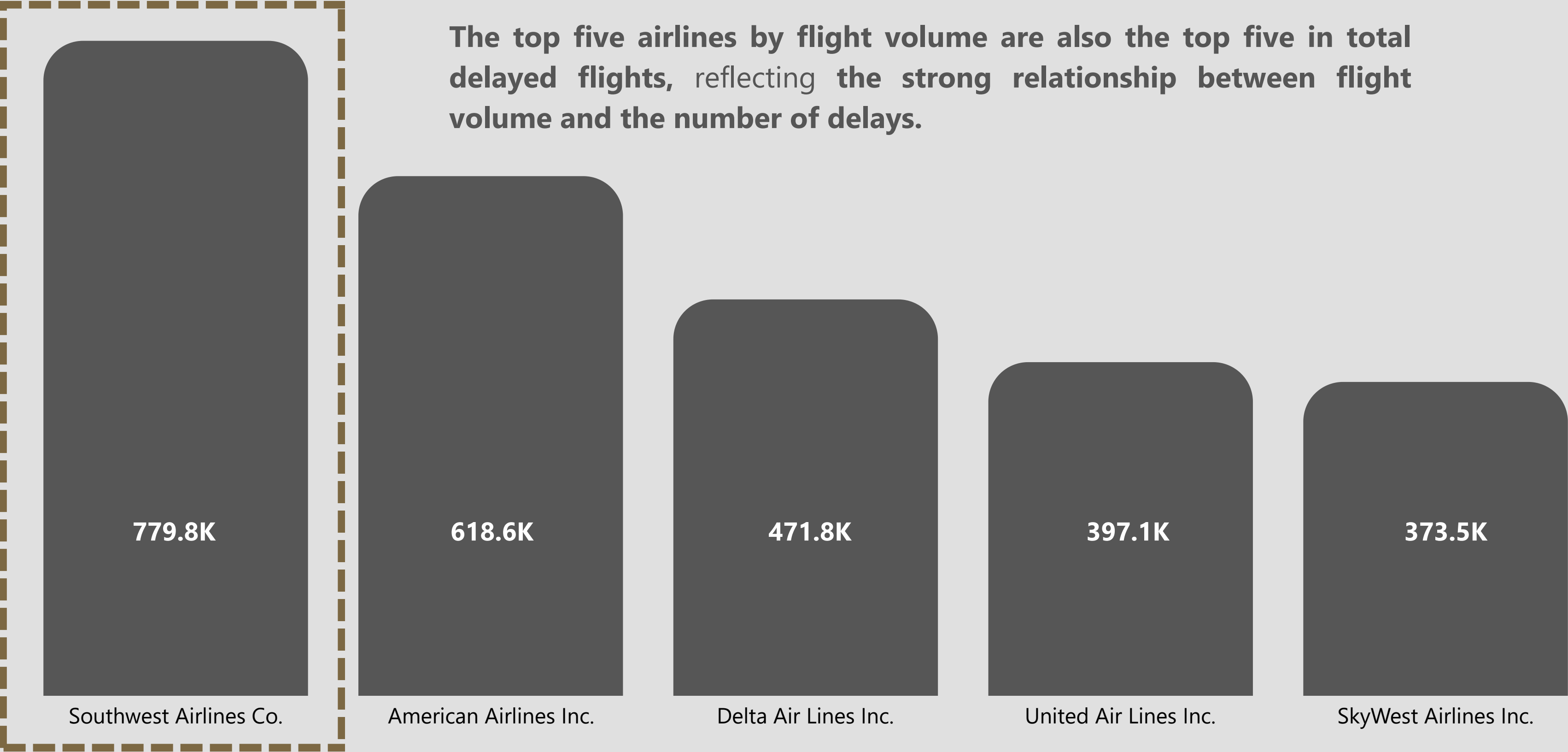


Number of Delays during the Year

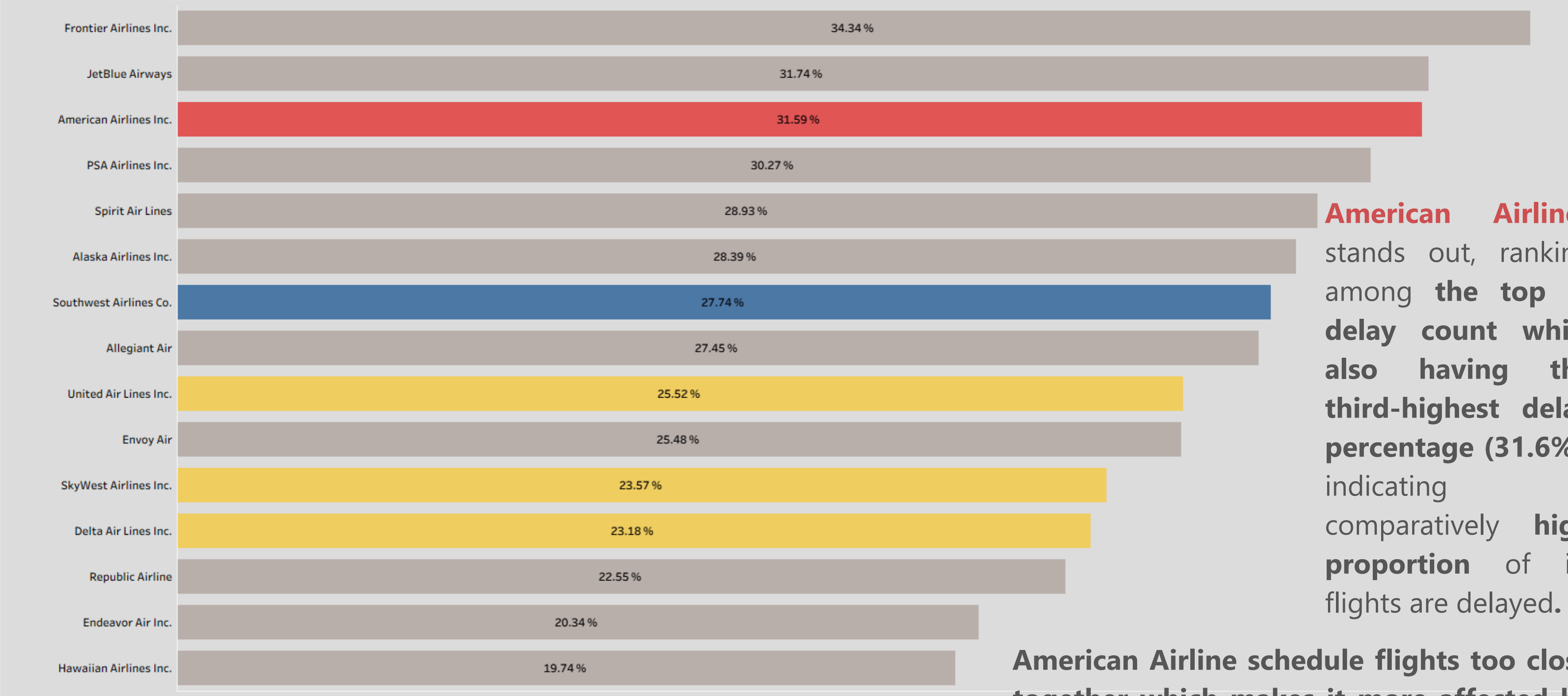


Top 5 Airlines by Number of Delayed Flights

The top five airlines by flight volume are also the top five in total delayed flights, reflecting the strong relationship between flight volume and the number of delays.



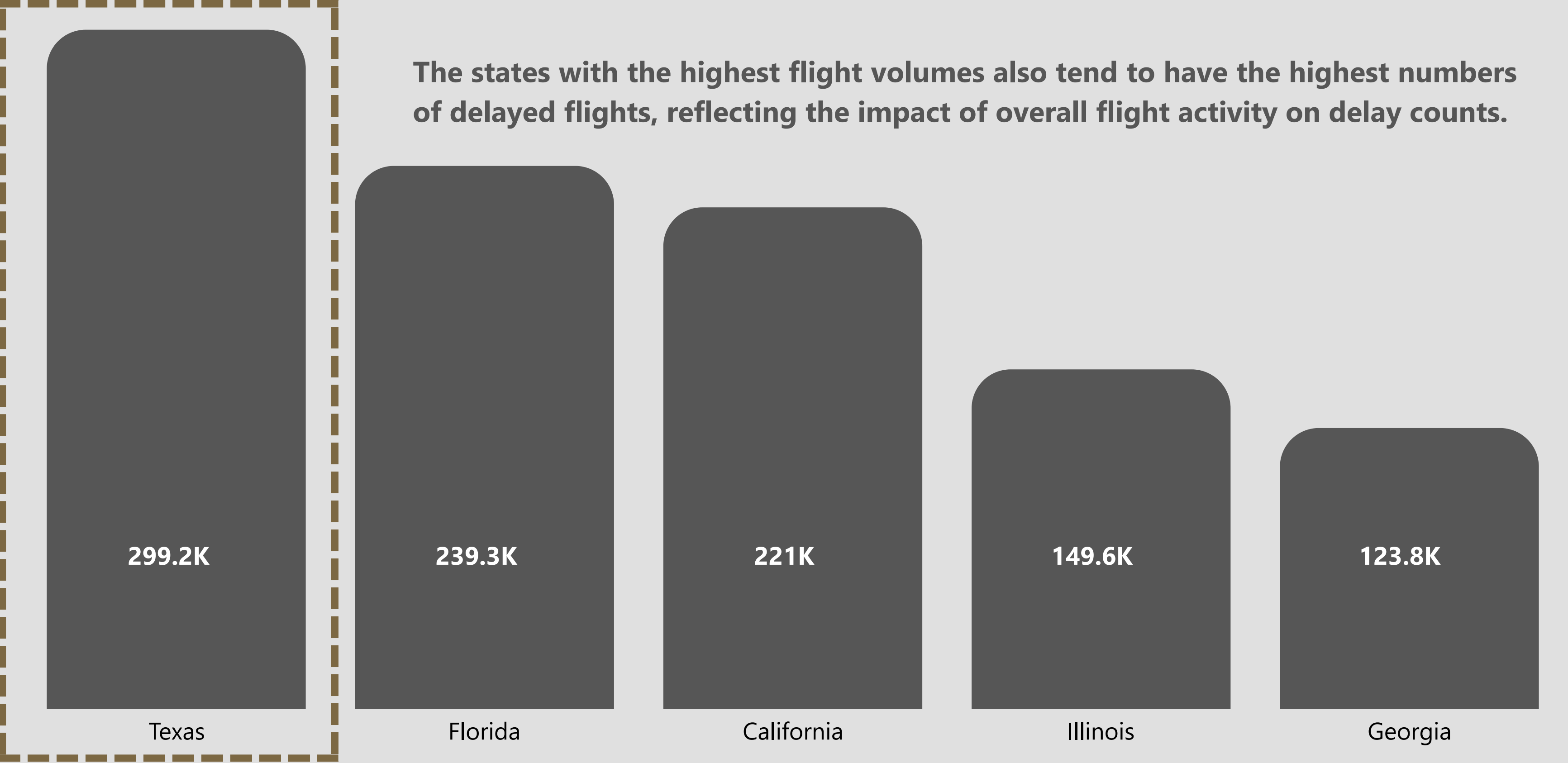
Airlines Delayed Flights Percentage



American Airlines stands out, ranking among the top in delay count while also having the third-highest delay percentage (31.6%), indicating a comparatively high proportion of its flights are delayed.

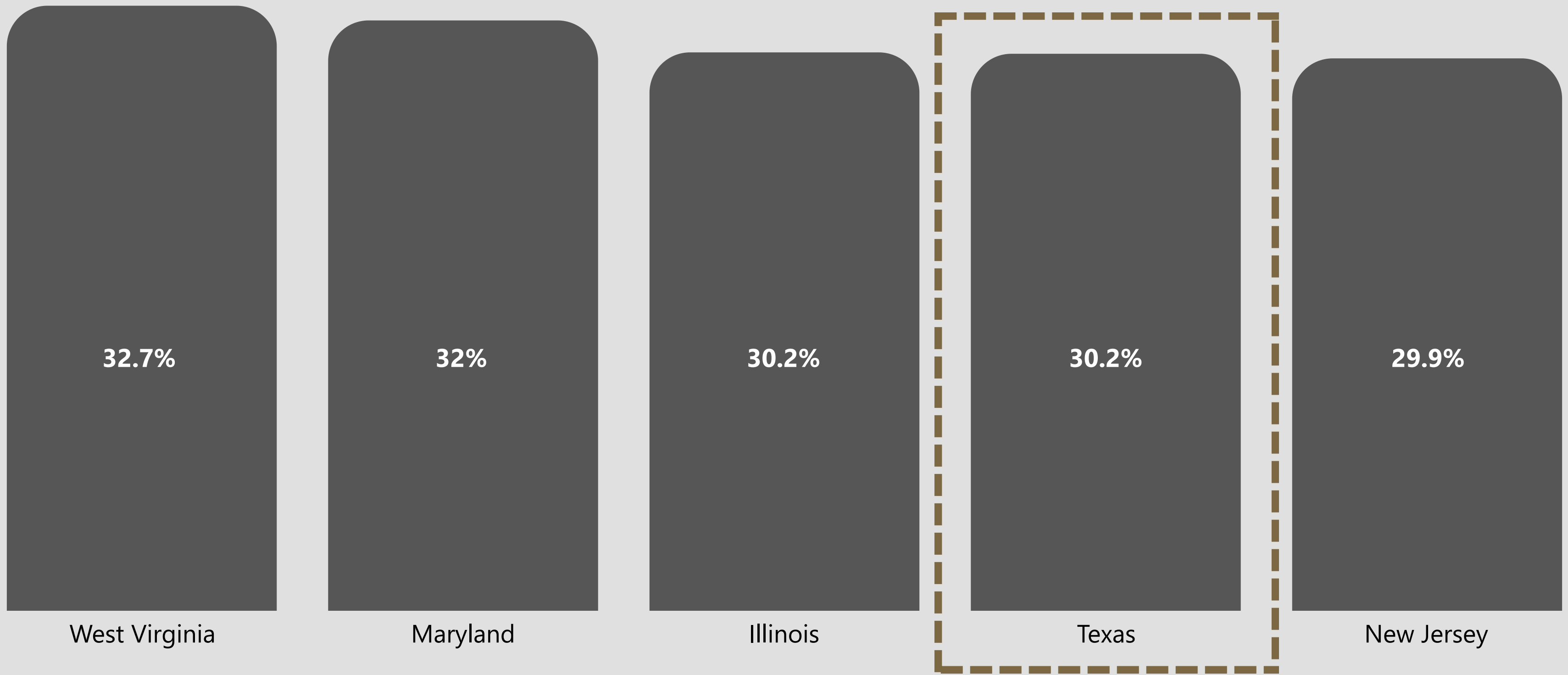
American Airline schedule flights too close together which makes it more affected by the domino affect of flight delays.

Top 5 States by Number of Delayed Flights



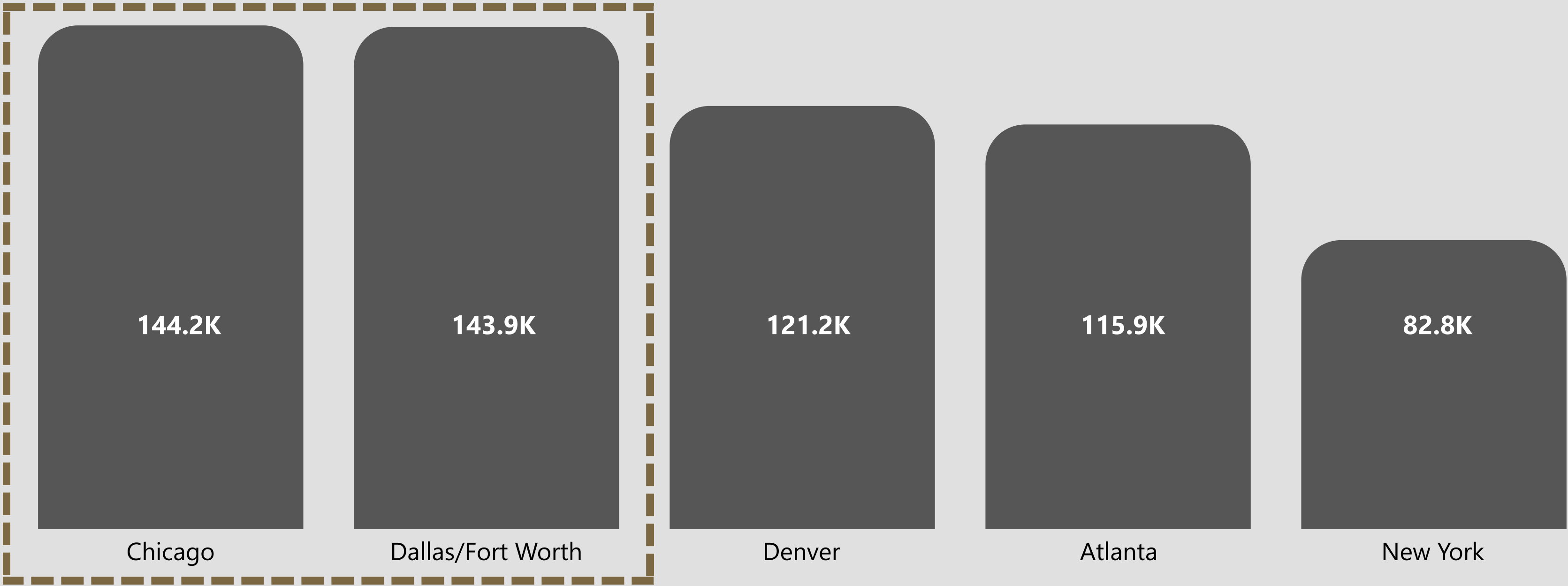
Top 5 States by Percentage of Delayed Flights

Although **Texas ranks first in total delayed flights**, it ranks **fourth in delay percentage**, showing that its high delay count is largely influenced by its high overall flight volume.



Top 5 Cities by Number of Delayed Flights

Chicago and Dallas/Fort Worth have the **highest number of delays** largely as both have **very high flight volumes**, meaning **congestion and operational disruptions** are affecting **a large number of flights**.



Big Airports

How were they defined?

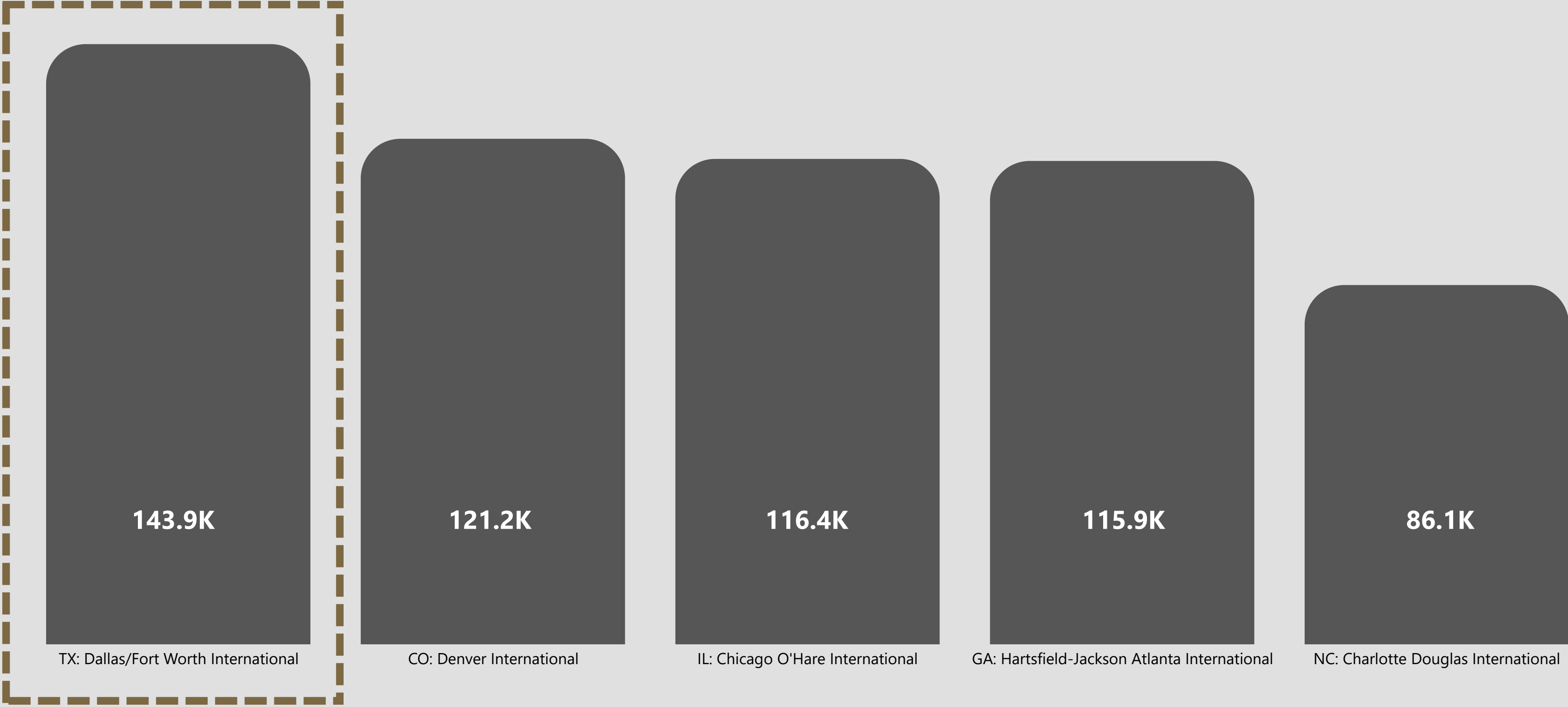


Flight Volume > 23K

23K flights = Median flight volume in International Airports

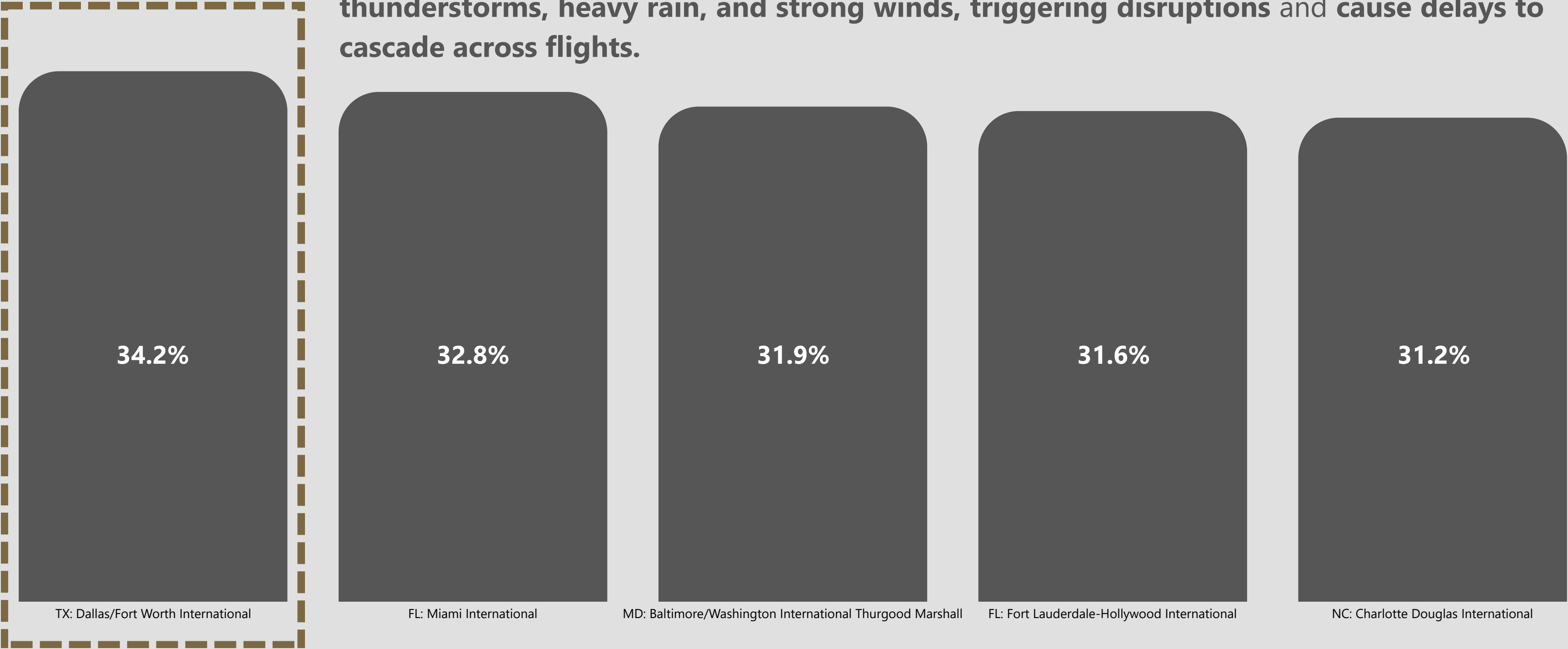
Airports with more than 23K recorded flights during the study period were classified as Big Airports and selected for deeper delay analysis.

Top 5 Big Airports by Number of Delayed Flights



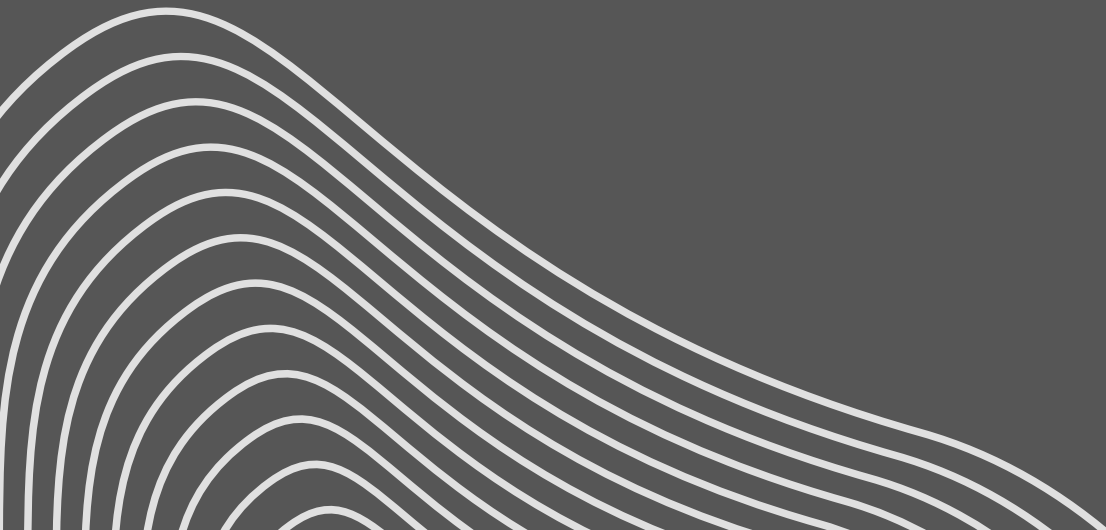
Top 5 Big Airports by Percentage of Delayed Flights

DFW has the highest delay volume due to its exceptionally high flight volume and major hub role for American Airlines. Its North Texas location also exposes operations to thunderstorms, heavy rain, and strong winds, triggering disruptions and cause delays to cascade across flights.





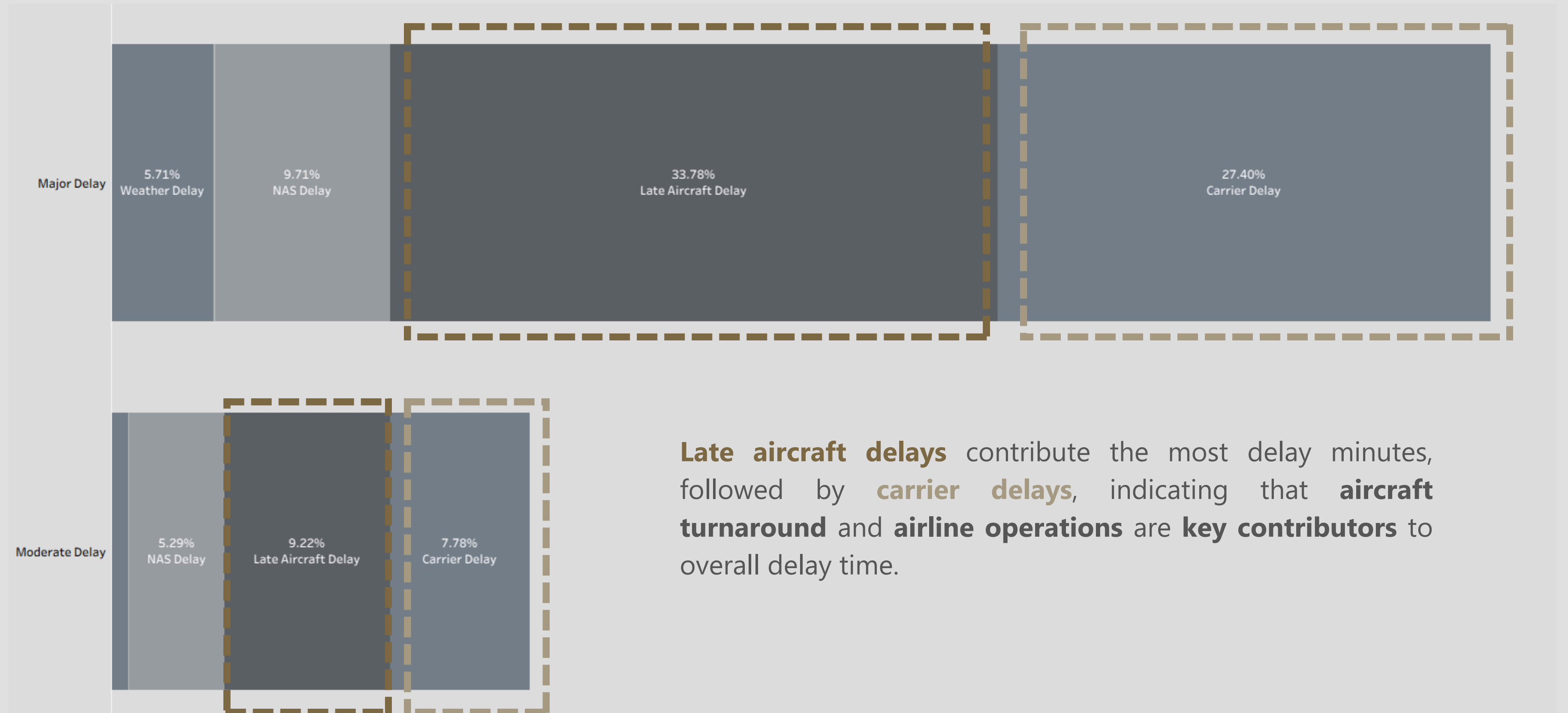
Delay Severity!



Number of Flights by Departure Delay Severity



Causes of Departure Delay per Delay Severity



Weather Delay per Month

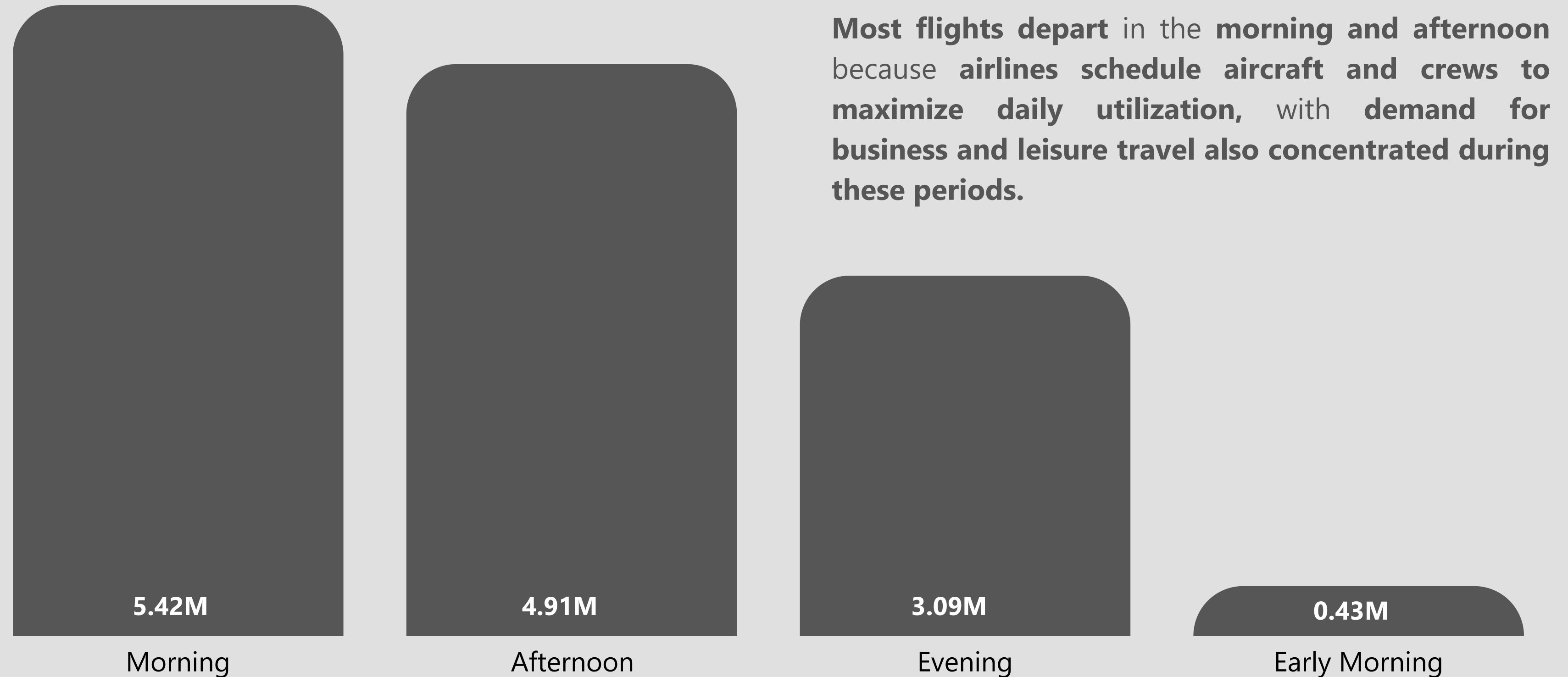
7.17

Weather-related delay minutes peak in **January** due to **winter conditions** such as **snow, ice,** and **reduced visibility,** while **October** generally experiences **milder weather and fewer disruptions.**

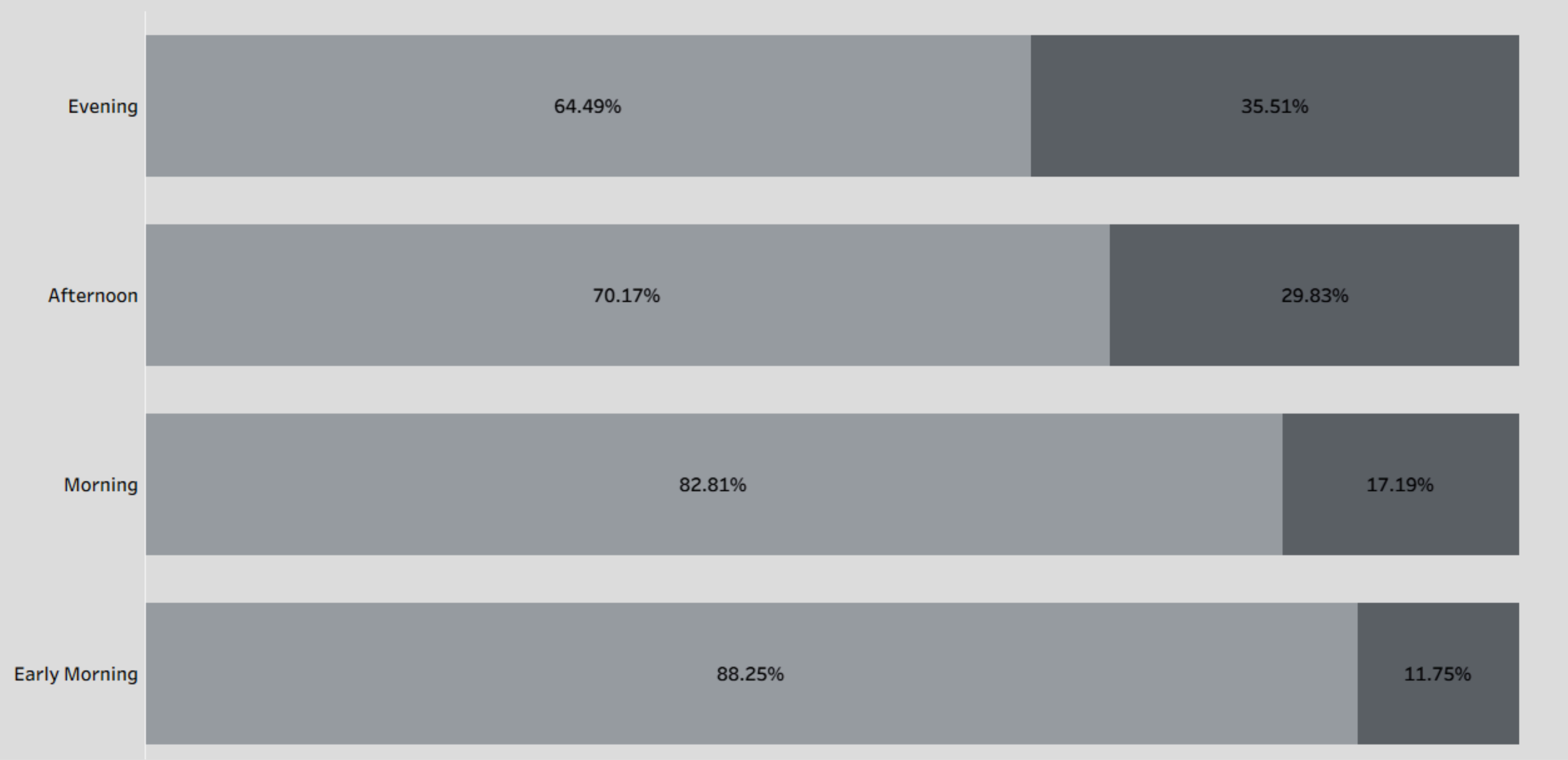
2.1

January February March April May June July August September October November December

Number of Flights per Departure Time

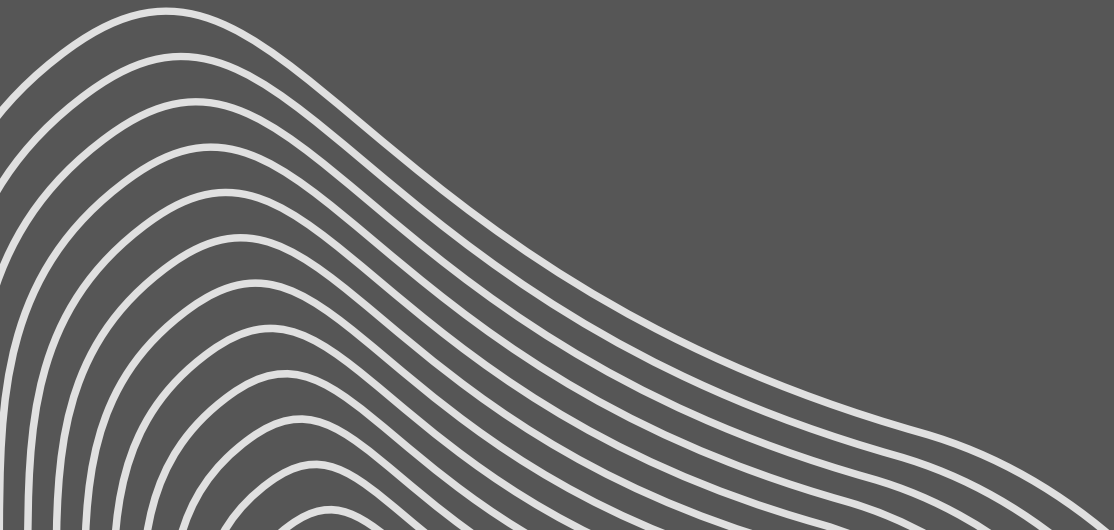


Percentage of Delayed Flights by Departure Time

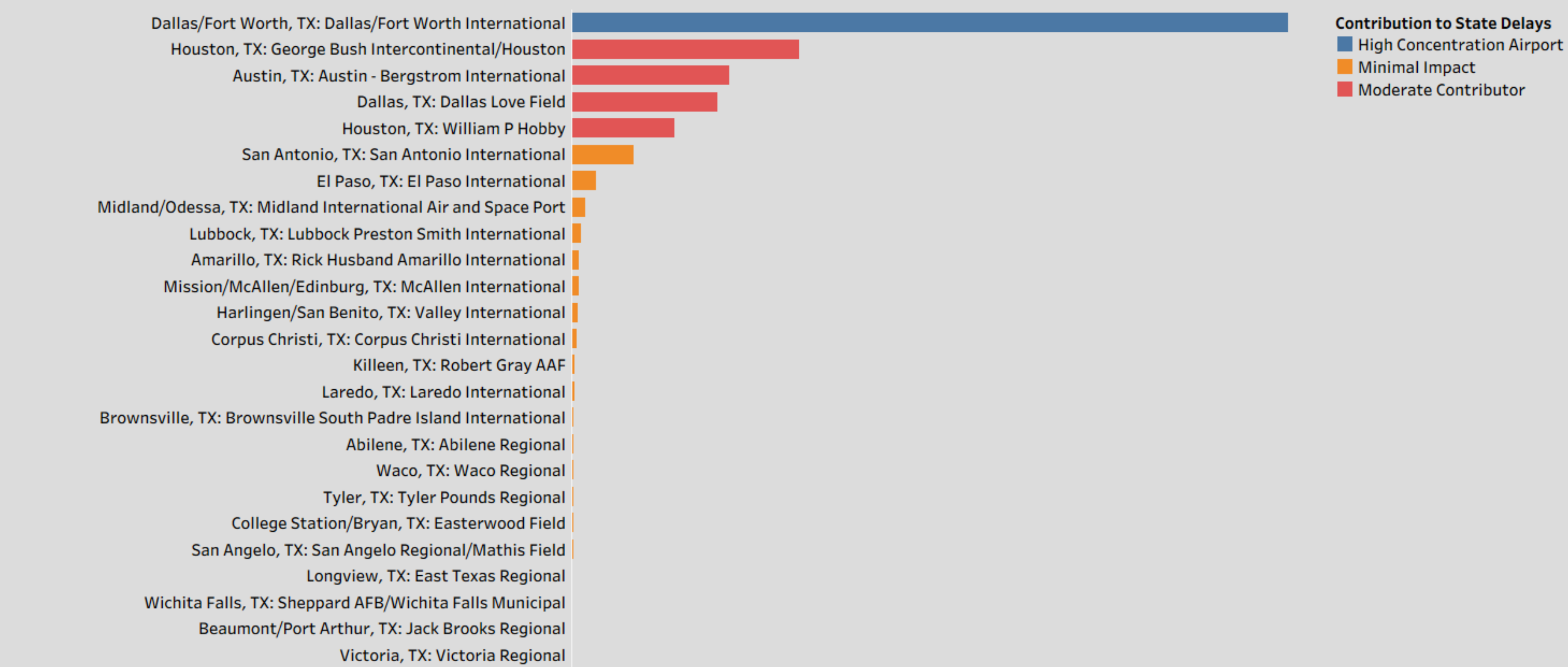


Evening departures tend to **have more delays** because **delays accumulated earlier in the day** can carry over through aircraft rotations and crew schedules, creating a cascading effect that becomes more noticeable later in the day.

Texas: Dallas/ Fort Worth International Airport



Delays Distribution across Texas Airports



DFW Airport

Located in North Texas

- **Highest delay percentage among major airports.**
- **3rd highest in total delay count among all airports.**
- **Highest contributor to total delays among Texas airports.**
- **7 runways — second-highest among U.S. airports, after O'Hare.**
- **3rd largest airport in the world by area and 2nd largest in the U.S.**



Airport Overview

Total Flights

629.44K

**Average
Departure Delay**

**22
Minutes**

Average Taxi In

**7
Minutes**

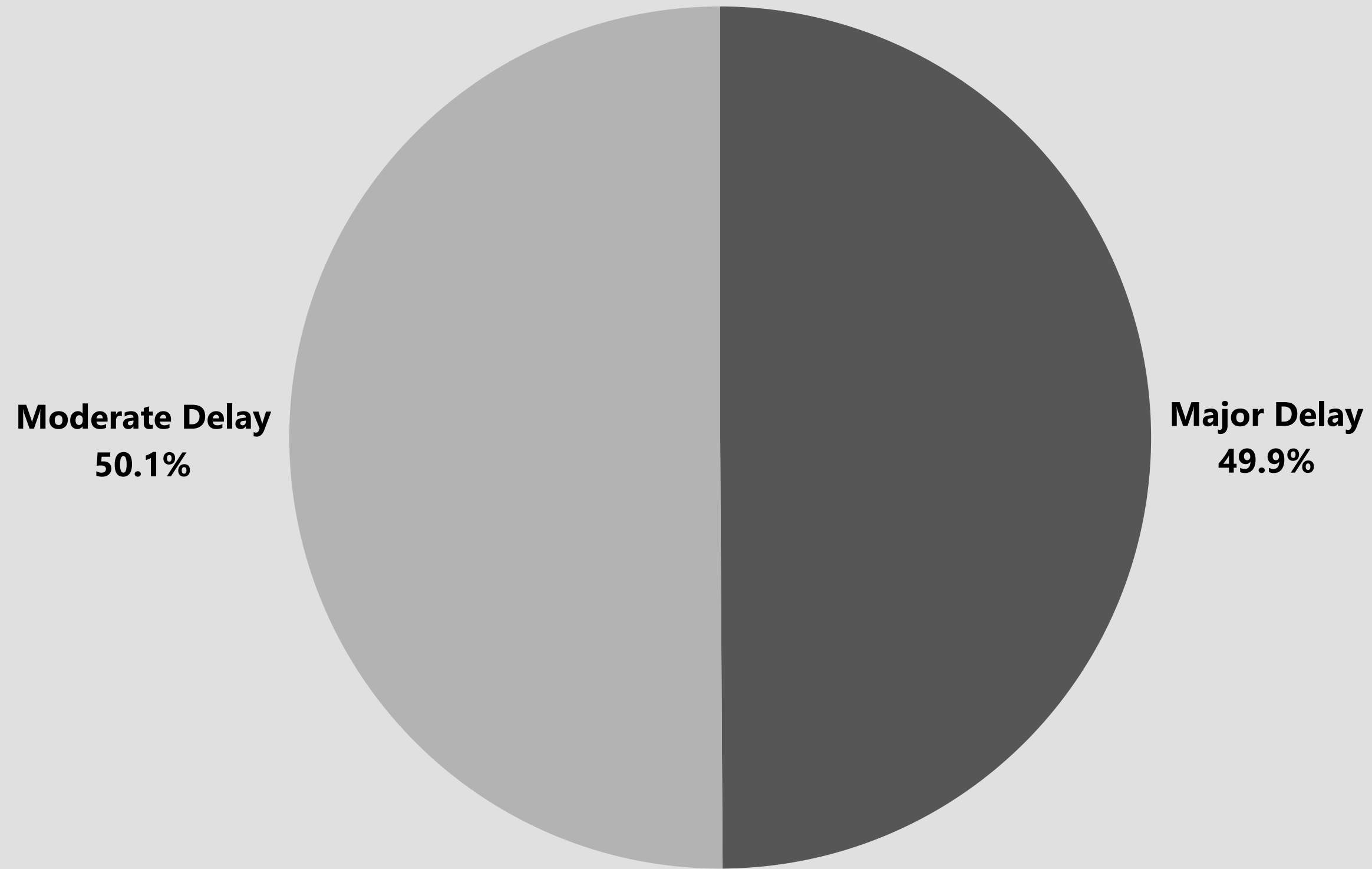
Average Taxi Out

**20
Minutes**

Number of Flights per Month from Airport



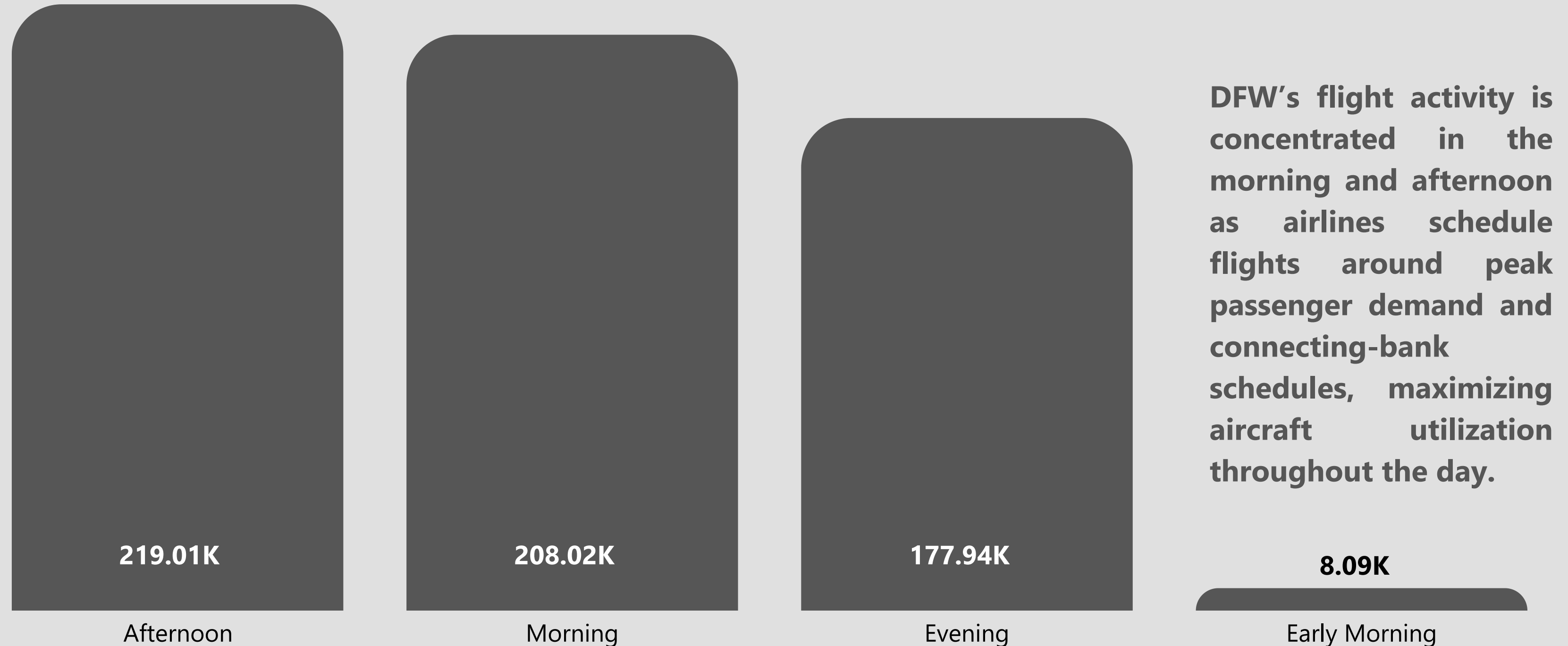
Delay Severity in Airport



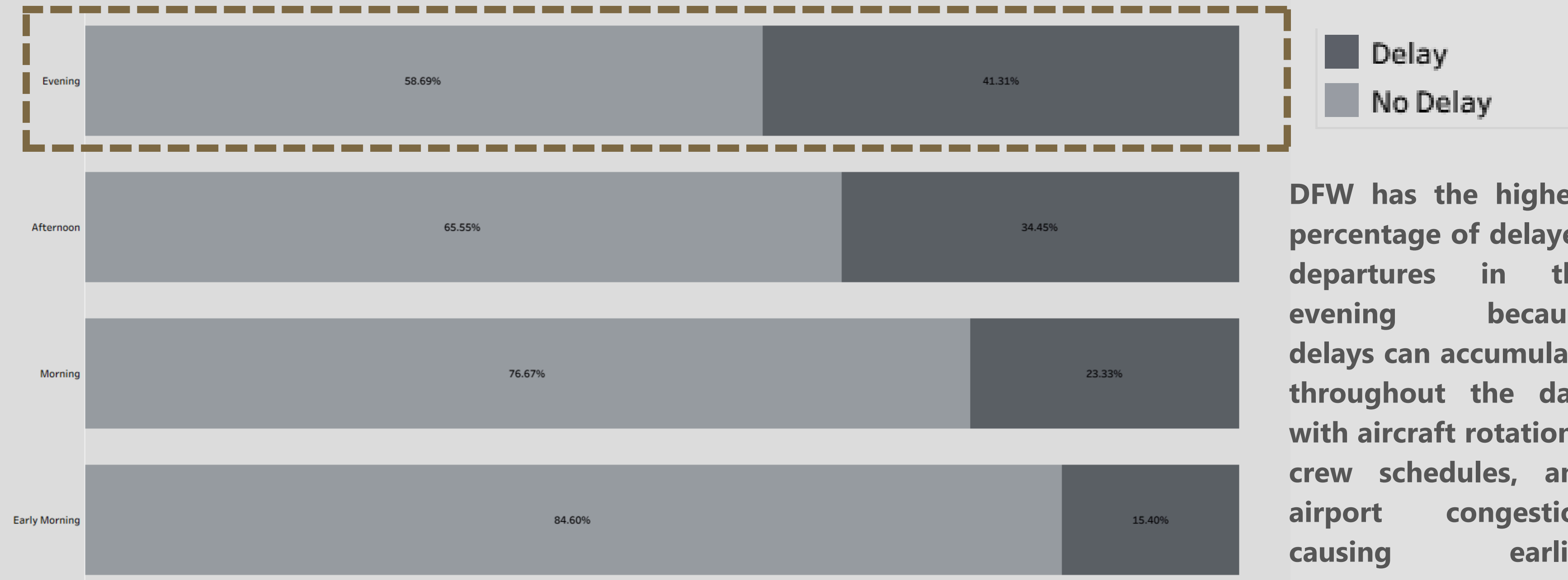
Top 5 Airlines by Number of Flights



Number of Flights by Departure Time



Percentage of Delayed Flights by Departure Time



DFW has the highest percentage of delayed departures in the evening because delays can accumulate throughout the day, with aircraft rotations, crew schedules, and airport congestion causing earlier disruptions to carry over into later flights.

Recommendations



Evening operations: Increase evening staffing and operational resources.



American Airlines: Improve aircraft turnaround, crew scheduling, and departure readiness.



Departure scheduling: Improve departure punctuality and add operational buffers.



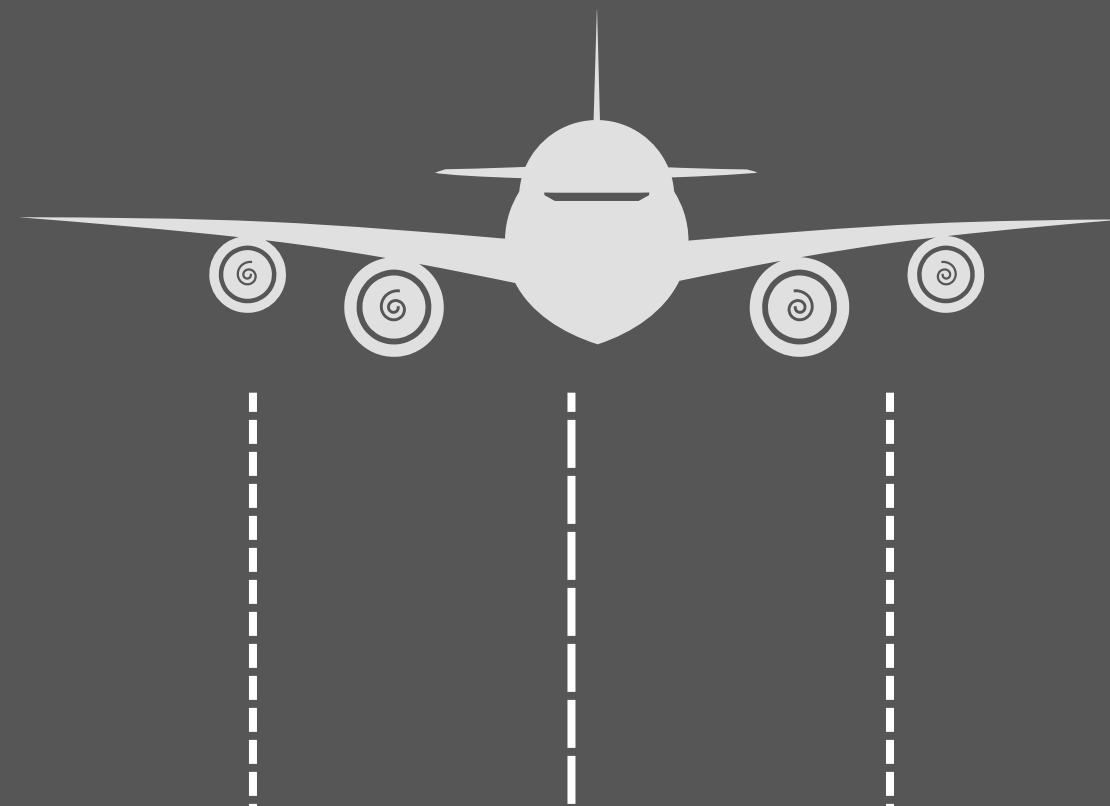
DFW Airport: Strengthen airport, airline, and ATC coordination during peak and weather disruptions.



Seasonal preparation: Increase resources and backup plan during peak seasons.



Dashboard



*Thank
You*

